

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing FINAL YEAR PROJECT INITIAL PROPOSAL

CrickSense

Project Team

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Project Proposal

Project Title: AI-Based Cricket Ball Tracking and Bowling Performance Analysis System

Proposed Application Name: CrickSense

1. Introduction

Cricket bowling performance depends on several important factors, including ball speed, swing, bowling line, bowling length, accuracy, and consistency. Traditionally, players depend on coaches or manual observation to understand these factors. Manual analysis can be time-consuming and may not always provide accurate or consistent measurements.

CrickSense is an AI-based cricket bowling analysis system designed to help players understand and improve their bowling performance. The system uses a camera to capture bowling deliveries and applies Computer Vision and Artificial Intelligence to detect and track the cricket ball.

The system analyzes each delivery and provides information such as ball speed, swing, bowling line, bowling length, pitch location, and ball trajectory. It can also analyze the relationship between the ball, batsman's pad, and wickets to provide a potential LBW assessment.

After analyzing a delivery, CrickSense provides a Delivery Quality Score and Wicket Threat Score. An AI-based analysis system then explains why a delivery was good or bad and provides recommendations to help the bowler improve.

The system also stores previous bowling sessions, allowing players to compare their performance and track their progress over time.

2. Problem Statement

Professional cricket teams use advanced technologies to analyze bowling performance, but these systems can be expensive and are generally not easily available to amateur and university-level players.

Developing players often do not have access to professional systems such as DRS and advanced LBW analysis because these technologies are expensive and mainly used in professional cricket. This limits their ability to receive detailed feedback and train effectively to improve their bowling performance.

Many developing bowlers depend on coaches or their own observation to understand problems with their bowling speed, line, length, swing, and consistency. This makes it difficult to accurately identify weaknesses and measure improvement.

3. Literature Review

Recent developments in Computer Vision and Artificial Intelligence have made it possible to analyze sports activities using cameras and machine-learning techniques.

Several cricket-analysis applications and systems provide automated analysis of bowling performance. Applications such as Matcha, Kricket.ai, CrickX, and Fulltrack AI provide different types of cricket analysis, including measurements related to bowling and ball movement.

Computer Vision techniques can be used to detect objects in video. Object detection models such as YOLO can identify objects such as a cricket ball, while tracking algorithms can follow the movement of the detected ball across consecutive video frames.

These technologies provide a suitable foundation for developing CrickSense. However, the proposed system aims to go beyond simply displaying measurements.

CrickSense will combine ball tracking and bowling measurements with performance scoring and AI-based feedback. This will help players understand why a particular delivery was effective or ineffective and how they can improve their bowling.

4. Proposed Solution

The system will use a smartphone camera placed at a fixed position behind the bowler or stumps. The recorded video will be processed using Computer Vision to detect and track the cricket ball.

The system will calculate:

- Ball speed
- Swing/deviation
- Bowling line
- Bowling length
- Pitch location
- Ball trajectory
- Bowling consistency
- Delivery Quality Score
- Wicket Threat Score
- Potential LBW analysis

An AI-based analysis layer will convert these measurements into simple feedback such as why a delivery was good or bad and how the bowler can improve.

5. Objectives

The main objectives are:

1. Detect and track the cricket ball using Computer Vision.
2. Estimate bowling speed and trajectory.
3. Analyze line, length, swing, and pitch location.
4. Provide bowling consistency and performance analysis.
5. Provide AI-based explanations and recommendations.
6. Develop a mobile application for viewing analysis and performance history.

6. Key Features

Core Features

- Cricket ball detection and tracking
- Ball speed estimation
- Swing/deviation analysis
- Line and length analysis
- Pitch map
- Delivery trajectory

Advanced Features

- Delivery Quality Score
- Wicket Threat Score
- Bowling Consistency Score
- Bowler Fingerprint/Profile
- Best Delivery Finder
- Automatic best-ball clip
- Potential LBW analysis
- AI bowling recommendations
- Session history and comparison

7. Scope of the Project

- Real-Time Video Processing
- Cricket Ball Tracking
- Bowling Performance Measurement
- Pad and Wicket Analysis
- Delivery Scoring
- AI-Based Feedback

- Bowler Performance Profile
- Session Management
- Performance Tracking
- Best Delivery Identification
- Automatic Reports

8. Expected Results

- Automated delivery-by-delivery bowling analysis.
- Bowling analysis using only a smartphone camera.
- Help players understand their bowling performance.
- Identify weaknesses in bowling.
- Compare performance between different sessions.
- Track and improve bowling consistency.

9. Limitations

The accuracy of the system may be affected by:

- Camera quality
- Camera position
- Lighting conditions
- Ball visibility
- Motion blur
- Ball speed
- Players blocking the ball
- Background conditions
- Limited camera angles

10. Conclusion

The proposed **CrickSense** system will combine Computer Vision, AI, and mobile application development to create an affordable cricket bowling analysis platform. The project will focus on automated ball tracking and meaningful performance feedback rather than simply displaying raw measurements.